

Sourabh Singh Rawat

Software Development Engineer Intern Candidate | Backend Development | Full-Stack Development | AI/ML Applications
Prayagraj, India | +91 8595591862 | sourabh.developer18@gmail.com
linkedin.com/in/sourabh-singh-rawat-321803369 | github.com/drawmebaaz

Professional Summary

Software Engineering student at IIIT Allahabad with hands-on experience developing full-stack applications, backend APIs, ML recommendation systems, and data-intensive analytics platforms using FastAPI, React, TypeScript, Node.js, REST APIs, WebSockets, PostgreSQL, MongoDB, Docker, JWT authentication, and Python. Strong foundation in Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks, OOP, System Design, Software Engineering, Machine Learning, and Probability.

Education

Indian Institute of Information Technology, Allahabad (IIIT Allahabad)

Bachelor of Technology in Information Technology - Business Informatics

2024-2028

CGPA: 7.94/10

Dr. B.R. Ambedkar School of Specialised Excellence, Kalkaji

Class XII: 95%

2024, New Delhi

St. Giri Public School

Class X: 96%

2022, New Delhi

Technical Skills

Languages: C++, Python, JavaScript, TypeScript, SQL, Java

Backend: FastAPI, Node.js, Express.js, REST APIs, GraphQL, WebSockets, Socket.IO, JWT Authentication, OAuth/OIDC, RBAC, Zod

Frontend: React.js, Streamlit, Tailwind CSS, Zustand, Vite, Recharts, Plotly, Responsive UI, State Management, Protected Routing

Databases: PostgreSQL, MongoDB, SQLite, SQLAlchemy ORM, Mongoose, Alembic, Schema Design, Indexing, Query Optimization

AI/ML and DevOps: pandas, NumPy, scikit-learn, Recommendation Systems, Feature Engineering, Monte Carlo Simulation, Docker, Docker Compose, Nginx, Render, Neon, Git, GitHub, Linux, pytest, Vitest, Ruff

Projects

AlgoRadar - AI Competitive Programming Analytics and Recommendation System

Python, Streamlit, pandas, NumPy, Plotly, scikit-learn, Codeforces API, LeetCode GraphQL, CodeChef Public Data

- Built a 3-platform competitive programming analytics system integrating Codeforces, LeetCode, and CodeChef profiles; analyzed up to 2,500 Codeforces submissions/user with cached profile and API responses.
- Engineered 9 solve-probability features from rating gap, problem rating, tag strength, solved volume, recent failures, popularity, recent accuracy, and calibrated platform difficulty.
- Designed a recommender generating up to 60 personalized practice problems/user across confidence, growth, and stretch buckets; delivered a 6-section Streamlit dashboard with 16 automated tests.

StockBreakers - Full-Stack Paper Trading and Quant Risk Analytics Platform

React.js, Tailwind CSS, Zustand, Recharts, Node.js, Express.js, MongoDB, Socket.IO, FastAPI, NumPy, Docker, JWT

- Architected a Dockerized 4-service paper-trading platform using React, Express/MongoDB, FastAPI analytics, and Nginx, supporting JWT auth, portfolio accounting, watchlists, order review, and virtual buy/sell workflows.
- Implemented real-time Socket.IO market streaming with simulated price updates every 4 seconds, synchronizing portfolio value, holdings, trade tickets, and market-watch state across the frontend.
- Developed quant analytics over 240-point rolling price history with up to 5,000 bootstrap Monte Carlo paths, 7-90 day forecast horizons, VaR/CVaR, downside probability, volatility, max drawdown, and Sharpe-style metrics.

Smart Hostel Grievance Intelligence Platform

React, TypeScript, FastAPI, Python, SQLAlchemy, Alembic, PostgreSQL, Docker, Tailwind CSS

- Launched a deployed full-stack hostel complaint platform with 2 role-based portals, 14 REST API endpoints, secure student/admin login, HTTP-only cookie authentication, RBAC, complaint history, and admin resolution workflows.
- Designed English and Hinglish complaint intake with hostel, location, category, and impact details; grouped related reports into priority-sorted hostel issues using 5 admin review filters.
- Structured the production FastAPI backend across 4 route groups and 5 database tables with SQLAlchemy models, Alembic migrations, service/repository layers, automated tests, Docker support, and Render/Neon deployment.

Relevant Coursework

Data Structures and Algorithms | Database Management Systems | Operating Systems | Computer Networks | Object-Oriented Programming | System Design | Software Engineering | Machine Learning | Probability and Statistics

Achievements

- Goldman Sachs Hackathon Round 1: Secured Rank 53 among approximately 16,000 participants.
- Competitive Programming: Codeforces Pupil, CodeChef 3 Star; solved 600+ DSA and algorithmic problems.
- AlgoUniversity Coding Contest: Ranked 2nd in a college-level competitive programming contest.
- WorldQuant International Quant Championship 2026: Stage 1 Certificate of Recognition; Top 20% of teams globally.
- JEE Main: Ranked among the Top 0.44% candidates nationwide.

Leadership and Activities

- Volunteer, IIIT Allahabad Campus Societies: Coordinated content preparation, creative planning, event support, and logistics across 3 societies: Sarasva Literary Society, Nirmiti Art Society, and Rangtarangini Drama Society.